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From Vision to Language: Investigating Causal Information Flow in Multimodal Decision-Making

Causal Patching of Video-Text Attention Reveals That VLMs Ground Vision at Answer Tokens, with Nouns as Anchors and Temporal Reasoning Fragile

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Vision-language models (VLMs) are widely deployed for video understanding tasks — question answering, visual reasoning, action recognition. Yet the internal mechanism by which visual information is integrated into language-based decisions remains largely opaque. This paper applies layer-wise causal intervention on video-text attention pathways to answer a basic question: at which layer, and at which token positions, does visual information causally determine the model's final answer? The answer is consistent across five VLMs spanning two orders of magnitude in parameters: visual information is integrated primarily when the model processes the candidate *answer* tokens, not the question. Nouns act as semantic anchors for cross-modal grounding; verbs carry temporal relational load; and temporal reasoning exhibits a distinct fragility suggesting models rely on linguistic cues rather than genuine frame-by-frame sequential understanding.

AT A GLANCE

Problem: Where and when do VLMs integrate visual signals when answering video questions?

Why it matters: Understanding integration sites guides architecture choices and reveals failure modes.

Method: Layer-wise causal patching of video-text attention activations; tested on spatial, causal, and temporal QA.

Result: Visual integration peaks at answer tokens; nouns anchor cross-modal binding; temporal reasoning fragility TFI=0.34.

Evidence: 5 VLMs (7B–72B); consistent across spatial, causal, and temporal tasks.

The Question

VLMs process videos as sequences of frame tokens and answer multiple-choice questions about their content. The prevailing assumption is that visual grounding is distributed throughout the forward pass. But mechanistic interpretability work on image-text models has suggested that cross-modal integration is often concentrated at specific layers and token positions. For video, with its additional temporal structure, the picture is even less clear.

This paper runs the first systematic causal intervention study on the cross-modal attention pathways of video VLMs, targeting three reasoning types: spatial (object location, spatial relations), causal (action outcomes, event causation), and temporal (sequence, duration, before/after).

Causal Patching Methodology

The central method is **causal activation patching** applied to cross-modal attention activations.

For a clean run on a video-question pair, all attention activations $\mathbf{A}_t^{(l)}$ at layer l, token position t are recorded. A corrupted run processes a corrupted video (shuffled or masked frames) to obtain degraded activations.

Causal patching restores the clean activation at a single (l^*, t^*) while all other activations remain corrupted:

$$\mathbf{A}_t^{(l)} = \begin{cases} \mathbf{A}_t^{(l)}[\text{clean}] & l = l^*,\ t = t^* \\ \mathbf{A}_t^{(l)}[\text{corrupted}] & \text{otherwise} \end{cases}$$

The **causal effect** of visual information at (l^*, t^*) is:

$$\text{CE}(l^*, t^*) = P(\text{correct} \mid \text{patched at } l^*, t^*) - P(\text{correct} \mid \text{fully corrupted}$$

High CE means visual information at that position is causally necessary.

Normalising by the maximum recoverable effect:

$$\widehat{\text{CE}}(l^*, t^*) = \frac{\text{CE}(l^*, t^*)}{P(\text{correct}|\text{clean}) - P(\text{correct}|\text{corrupted})}$$

Where Visual Integration Happens

The results are striking in their consistency. Across all five models and all three task types, the highest causal effects appear at the positions of *candidate answer tokens* — not the question, not the reasoning prefix. The model reads the video primarily when it evaluates each answer option against the video content.

	Spatial	Causal	Temporal
Peak CE at answer tokens	0.71	0.68	0.53
Peak CE at question tokens	0.18	0.22	0.19

Visual integration peaks at layers 60–75% of model depth, in the middle-to-late transformer layers.

Nouns Anchor, Verbs Carry Temporal Load

Aggregating causal effects by token part-of-speech reveals the role of different word classes:

$$\widehat{\mathrm{CE}}_{\mathrm{POS}}(\mathrm{pos}) = \frac{1}{|T_{\mathrm{pos}}|} \sum_{t \in T_{\mathrm{pos}}} \sum_{l} \widehat{\mathrm{CE}}(l, t)$$

POS	Spatial	Causal	Temporal
Noun share	58%	51%	44%
Verb share	12%	24%	41%
Other share	30%	25%	15%

Nouns dominate as binding sites for object identity and spatial information. Verbs grow in importance for causal and temporal tasks, where relational and sequential information must be extracted from the video.

Temporal Fragility

The **Temporal Fragility Index** (TFI) measures how much harder it is to recover visual information for temporal questions:

$$\mathrm{TFI} = \frac{\widehat{\mathrm{CE}}_{\mathrm{spatial}} - \widehat{\mathrm{CE}}_{\mathrm{temporal}}}{\widehat{\mathrm{CE}}_{\mathrm{spatial}}}$$

TFI $= 0.34$ averaged across models. Restoring a clean visual signal for a temporal question recovers 34% less accuracy than for a spatial question, despite identical video content.

Combined with the high verb CE share on temporal tasks, this pattern suggests VLMs use linguistic temporal markers (verb tense, temporal adverbs, "before/after" phrases) rather than reconstructing sequential information across frames. Temporal expressions in answer candidates trigger reliance on linguistic temporal priors.

Experimental Details

Models: LLaVA-Video-7B, LLaVA-Video-72B, Qwen2.5-VL-7B, Qwen2.5-VL-72B, InternVL3-8B.

Video benchmarks: Spatial, causal, and temporal questions derived from standardised evaluation datasets; 4 answer candidates per question; balanced correct/incorrect.

Corruption: Frame shuffling and frame masking produce similar CE patterns, confirming that ordering information (not just content presence) drives temporal fragility.

Model scale: Larger models achieve higher absolute accuracy but the same qualitative pattern, suggesting these are architectural properties rather than capacity limitations.

Reference

Davide Testa, Hugh Mee Wong, Alessandro Lenci, Bernardo Magnini, and Albert Gatt. **From Vision to Language: Investigating Causal Information Flow in Multimodal Decision-Making.** arXiv:2609.05149, September 2026. https://arxiv.org/abs/2609.05149